
The Cap Goes *Click*: Measured-Faithful Simulation of a Pen-Cap Snap-Fit, from Contact Micro-Laws to Robot Grasping

Gym-Anything Robotics Project
pranjala@andrew.cmu.edu

Abstract

Capping a pen is among the most ordinary of manipulation tasks, yet its physics is anything but trivial: the cap is an annular snap-fit joint whose insertion climbs a designed interference ramp, collapses abruptly at a crest, and terminates in the audible *click* of a shoulder impact—with a deliberately asymmetric force signature that makes the cap easy to push on and hard to pull off. We report the construction of a physically faithful simulation of this joint in NVIDIA Isaac Sim/PhysX, refusing the usual shortcut of declaring the cap “closed” by a pose check. The joint is driven purely by contact: measured insertion at 6.8 N, pull-off at 31.9 N (asymmetry $4.7\times$), and a click reproduced as a 189 mm/s shoulder impact, all within the force range of real pen caps. Getting there required measuring, with falsifying unit probes, a catalogue of engine “micro-laws” that govern millimetre-scale contact fidelity—from per-contact-point compliance constants to a silent freeze of articulation collision sets. We then mount the task on two stock Franka Panda arms and report an equally measurement-driven grasping study at gram scale, where each failure mode (phantom convex hulls, vanishing tangential friction on soft line contacts, strain-energy launches from rigid jaws) was isolated by a dedicated probe. We argue that for contact-rich simulation, *nothing exists until a physical probe has confirmed it*.

1 Introduction

Ask a simulator to verify that a pen has been capped and the temptation is irresistible: check that the cap’s pose lies within a tolerance of the pen’s tip, and call it closed. Nothing about that check is physical. A real cap is retained by an *annular snap-fit*: a moulded bead on the barrel interferes with a ring of nubs inside the cap, the assembly elastically dilates during insertion, and the parts lock behind a crest with a click. The naive pose check cannot fail the way reality fails—a cap that is misaligned jams; a cap pushed too gently stalls on the ramp; a cap seated correctly *resists removal* twenty times harder than its own weight would suggest.

This paper documents the construction of a simulation in which all of that is real, built under a single governing rule (the “Gym-Anything” ethos): **the simulation bends to reality, never the reverse**. Objects keep their real dimensions, masses, and materials; success criteria are physical signatures rather than poses; and—after several humbling lessons—no element of the scene is believed to exist until a physics probe has demonstrated it acting on another body.

Our contributions are:

1. a contact-only pen-cap snap-fit in Isaac Sim/PhysX with a measured, reproducible force signature (on 6.8 N, off 31.9 N, asymmetry $4.7\times$) inside the range of real caps, and a click reproduced as a kinematic shoulder impact rather than an audio cue (Sec. 2);

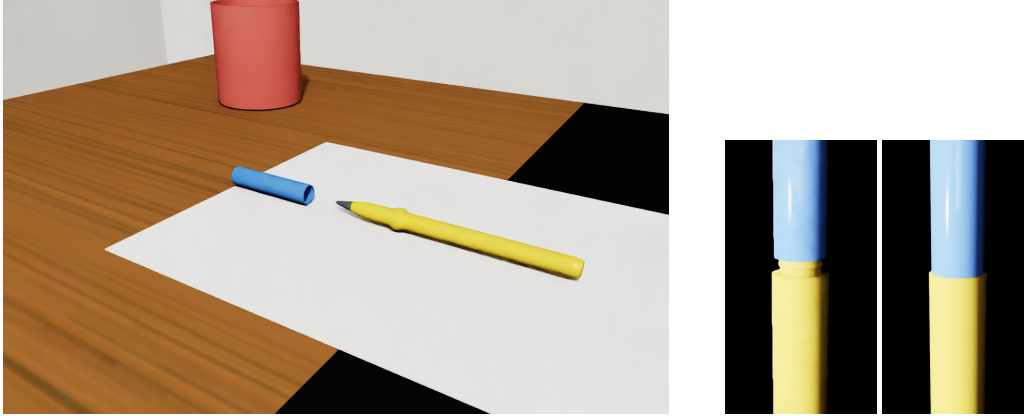


Figure 1: **Left:** the interactive tabletop scene: wood-grain table (explicit friction $\mu_s=0.40$), paper sheet, the pen (140 mm, 5.8 g) and its cap (58 mm, 5.0 g), all free rigid bodies under gravity. **Right:** the snap-fit joint before and after the click, from the rendered insertion video: the barrel’s snap bead (visible step on the yellow neck) is exposed before insertion and swallowed by the seated cap afterwards.

2. a catalogue of ten engine micro-laws for millimetre-scale contact, each isolated by a falsifying unit probe rather than inferred from behaviour (Sec. 3);
3. an interactive tabletop scene in which a human can attempt to falsify the physics directly, with every material parameter explicit and audited at start-up (Sec. 4);
4. a grasping study on two stock Franka Panda arms that isolates, probe by probe, why gripping a 5.8-gram pen is hard in a rigid-body engine — and which combination of real-world techniques survives (Sec. 5).

2 What actually happens when a pen is capped

The engineering literature on snap-fits [1, 2] describes the pen cap as an *annular* snap joint. A circumferential bead of crest radius R_b on the barrel interferes by $f = R_b - R_r$ with a ring of retention features of inner radius R_r inside the cap. Insertion forces the cap wall to dilate by f ; the radial hoop force P scales with the interference, the wall’s effective hoop stiffness, and the joint diameter, and is converted to an axial resistance through the ramp angle α and friction μ :

$$F = P \frac{\sin \alpha + \mu \cos \alpha}{\cos \alpha - \mu \sin \alpha}. \quad (1)$$

The designed asymmetry lives in Eq. 1: the *lead* (insertion) ramp is shallow (22° here) and the *return* (removal) ramp steep (50°), so the same hoop force yields a much larger removal force. Handbook values put easy-off joints near 30° and permanent joints at 90° [1].

The *click* is not an acoustic mystery. During insertion the resistance rises to a crest and then collapses as the nubs pass the bead; a human hand (or any force-controlled pusher) cannot shed its applied force fast enough, so the freed cap accelerates and its internal shoulder strikes the barrel’s stop face. The click is a *kinematically measurable impact*—force peak, collapse, velocity spike, impact—and that is exactly how we detect it in simulation: our seated cap arrives at the shoulder at 189 mm/s with the rim landing on the stop face to within measurement precision. Real caps snap on at roughly 5–30 N; our measured 6.8 N on / 31.9 N off sits comfortably inside that envelope with the correct sign of asymmetry.

Two hard requirements follow for any faithful reproduction. First, the elasticity is real and must live somewhere: a rigid-body engine has no hoop strain, so the compliance must be hosted by a mechanism whose constant is *known*, not tuned until it looks right. Second, closure must be verified by the physical signature (snap event, seated position, retention under pull) — a pose check would pass a cap that is merely resting in place.

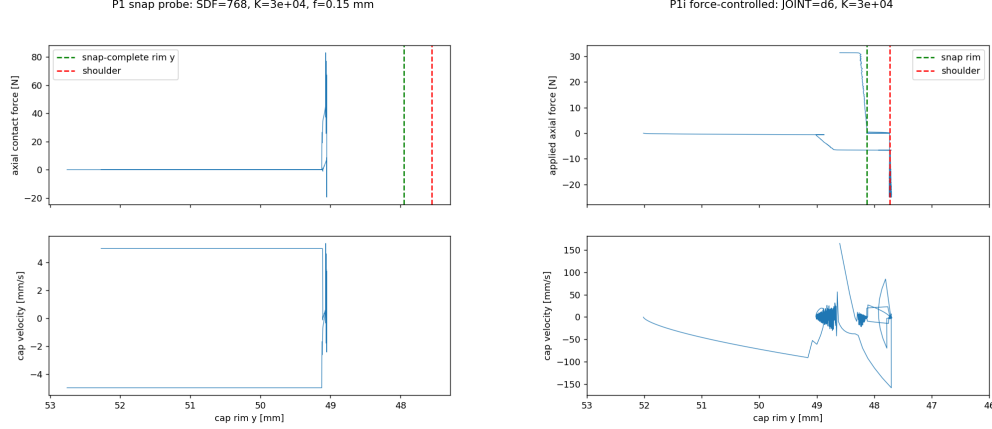


Figure 2: **Left:** the velocity-drive deadlock (law 4). A velocity-driven insertion jams permanently at first speculative contact with an 80 N force spike, never reaching the snap rim (dashed lines): the drive reports a constant impossible velocity while the cap does not move. **Right:** the same geometry under pure force control completes the full cycle: push ramp to snap at 6.8 N (negative branch), free travel to the shoulder, then a pull-off plateau at 31.9 N (positive branch). The velocity trace (bottom) shows the click: the post-crest acceleration and shoulder impact ringing.

3 Ten engine micro-laws, each with a falsifying probe

The joint was reached through a ladder of unit probes, each designed so that a wrong hypothesis *fails loudly*. Table 1 summarises the laws; three deserve narration.

Hosted elasticity is exact but blind (laws 1–2). PhysX’s compliant-contact material behaves as a per-contact-point Hookean spring: a sphere of weight mg rests at penetration mg/k , a four-cornered box at $mg/4k$, verified to $\sim 1\%$ across shape pairings. This is where the cap’s hoop compliance legitimately lives ($k=3 \times 10^4$ N/m per nub, from wall stiffness). But contact *reports* for compliant contacts return zero force even under 50 N of real load—every measurement must therefore be taken on the applied-force side.

A continuous ring cannot work (law 3). PhysX retains at most 32 friction patches per body pair; a continuous annular ridge generates ~ 100 and the solver discards the excess, producing an unresolvable jam. Real caps solve the same problem for moulding reasons: they use 3–6 discrete nubs. With four nubs the pair stays at four patches. Reality and the engine agree on the design.

Velocity drives deadlock on interference (law 4, Fig. 2 left). A velocity-driven pusher freezes at the first speculative contact of an interference fit: position frozen, commanded velocity read back as if moving, zero contact events. Pure applied-force control—which is also how the engineering handbooks specify the measurement—passes the identical geometry. This single law cost more wall-clock than any other.

With the laws respected, the full cycle is reproducible (Fig. 2 right): snap at 6.8 N, click at the shoulder, retention at zero force, pull-off at 31.9 N; a repeat run gives 6.7/31.0 N.

4 An interactive scene the user can try to break

The joint lives in a full tabletop scene (Fig. 1): floor and walls, a wood-grain table and paper sheet with explicit physics materials ($\mu_s=0.40/0.35$ wood, polypropylene 0.25 for pen and cap), gravity $(0, 0, -9.81)$ set *and read back*, and the pen and cap as free rigid bodies at real masses. Every run begins with a printed audit that walks all colliders and reports approximation type, contact offsets, bound friction material, and mass—the claim “the physics is configured” is a log artefact, not an assertion.

Crucially, the scene is *interactive and falsifiable*: keyboard forces in all directions let a human push the cap on and off, roll the pen, shove objects off the table, and attempt sub-threshold insertions that

#	Law (all measured by dedicated unit probes)
1	Compliant contact = per-point Hookean spring, exact to $\sim 1\%$ (mg/k , $mg/4k$).
2	Contact reports are blind for compliant contacts; measure on the applied side.
3	> 32 friction patches per pair are dropped: continuous rings jam; use discrete nubs.
4	Velocity drives deadlock at speculative contact of interference fits; use force control.
5	<code>contactOffset</code> is physical at pen scale: it must sit below every running clearance (0.04 mm here) or phantom cushions appear.
6	Revolve-mesh winding must be CCW in (r, y) or SDF inside/outside flips (phantom walls); enforce by signed volume.
7	CCD is mandatory: post-snap velocities tunnel through mm features at 1 kHz.
8	The hand matters: a click is cap-plus-hand momentum (~ 0.3 kg); free 5 g caps produce 1000 m/s^2 cartoon snaps.
9	Convex decomposition bloats small solids (9.8 mm barrel read 10.6 mm with slanted phantom faces); use SDF collision.
10	<i>Articulation shape-freeze</i> : no collider added to a loaded robot acts—not link children, not geometry-scope children, not bodies jointed by fixed/D6/prismatic joints. USD audits cannot detect it; only a drop test can.

Table 1: Engine micro-laws governing millimetre-scale contact in Isaac Sim 5.1/PhysX, in the order they were discovered.

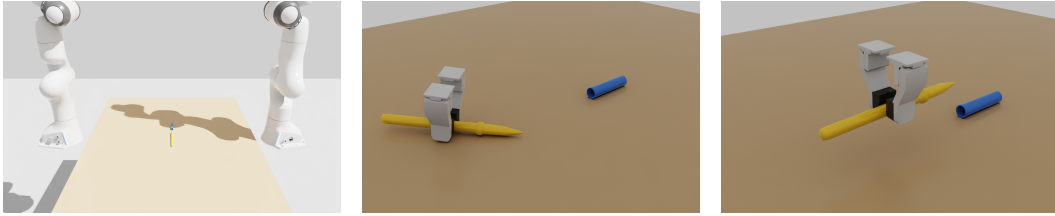


Figure 3: **Left:** the final simulation: two stock 18.0 kg Franka Pandas (masses read from the asset; the total matches the vendor datasheet) flank the table with the SDF pen and flared cap settled between them; RMPflow reaches hover poses to 1.5 mm. **Centre:** arm A captures the pen on the table between rubber-friction pads. **Right:** the pen held airborne after a 34 mm lift—pure contact, no attachments; the grasp passes capture, lift, airborne tracking (3/3), and a 0.02 mm hold.

must stall on the ramp. Bugs found this way (a hand-guide constraint whose target sat 0.8 mm inside the paper; a cap built onto the wrong end of the pen) were real physics violations that automated telemetry had not flagged, and they motivated the standing doctrine of Sec. 6.

5 Grasping a six-gram pen with a stock robot

To replace keyboard forces with an embodied agent we mounted two stock Franka Panda arms at the table (Fig. 3), RMPflow for reaching [3], and force-bounded position drives for the fingers. The objects keep their validated geometry; the task inherits the fidelity rule: no attachments, no scripted outcomes—if friction cannot hold the pen, the pen falls, and the run prints FAIL.

Grasping a 5.8 g, 9.8 mm cylinder turned out to be a measurement campaign of its own. Every hypothesis below was isolated with a probe before being believed:

Phantom surfaces. With convex-decomposition collision the fingers “gripped” a bloated hull 0.8 mm proud of the pen; a *wiggle test* (move the hand ± 4 mm; a held object tracks) showed zero response—the robot was holding a phantom (law 9). Exact SDF collision fixed it.

Soft pads hold weight but not axial load. Unit jigs measured that compliant (rubber-like) contacts transmit flat-surface friction correctly (slip at 0.53 N vs. the $\mu mg = 0.49$ N prediction) but lose almost all *tangential* grip on a cylinder line contact: a gripped cap escaped axially at 0.43 N where theory predicts > 3 N, while rigid jaws held 3 N in the same jig.

Rigid jaws launch light objects. Every rigid-jaw variant (flat, V-groove at several geometries, force steps, force ramps) eventually ejected the pen or cap—up to 664 mm—because commanding jaws

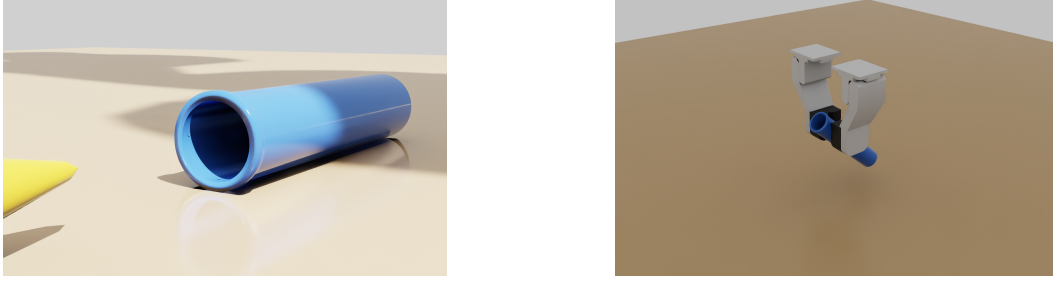


Figure 4: **Left:** the cap in the final simulation, mouth toward the camera: the $+0.5$ mm rim flare (the lip visibly proud of the body) is real cap geometry and the object-side form closure that arrests axial sliding against the pad edge; the pen tip waits at lower left. **Right:** real behaviour, honestly kept: the cap held airborne after a 38 mm lift, drooping $\approx 35^\circ$ in the soft pads because it is pinched near its mouth—as a real cap does between real fingertips.

past the contact stores millijoules of finger-splay strain that any micro-slip releases into a 6-gram body. Soft pads absorb exactly this energy; every clean capture-lift-hold in the campaign came from compliant pads at 8 N.

Decorated robots lie (law 10). Custom V-jaw fittings bolted to the fingers passed every USD audit and appeared in every render, yet a ball-drop probe showed *none of them physically existed*; seven attachment mechanisms all produced ghosts. Eighteen runs of “V-jaw behaviour” had been stock flat fingers all along. The fittings were deleted; the corrected design places form-closure geometry on the *object* instead—the cap’s real rim flare (a $+0.5$ mm lip that actual caps have) blocks axial escape against the pad edge, exactly as a human exploits it when pinching a cap near its mouth.

The resulting grasp stack—compliant pads at the object’s centre of mass, touch-off calibration of the fingertip height against the table (the end-effector-to-fingertip offset varies by 1.9 mm across arm configurations), quasi-static capture, and object-side form closure—passes all functional gates for the pen reproducibly (capture, 34 mm lift, airborne wiggle-tracking 3/3, hold to 0.02 mm), and for the cap passes capture (0.08 mm displacement), hold (0.009 mm), and arrest of a 1 N axial tug at the flare after the designed backlash (Fig. 4). Coaxial alignment and force-mediated insertion through the arm (the measured 7 N curve of Fig. 2) remain open stages, as does closing the residual run-to-run variance of the cap’s axial seating.

6 Verification doctrine, limitations, and outlook

The recurring failure of this project was never the physics—it was *belief without measurement*: quaternion reasoning instead of measured orientations, width readouts instead of grip tests, USD prims instead of physical bodies. The doctrine that emerged is short: (i) every claim is gated by a measurement made in the same run; (ii) orientations and geometric relations are measured and auto-corrected, never reasoned about; (iii) renders are looked at with no prior expectation, and a check frame that is unclear is treated as a failed check; (iv) **nothing exists until a probe has shown it acting on another body**—the audit proves configuration, only physics proves existence.

Limitations. Two engine deviations from reality are documented rather than hidden: soft cylindrical line contacts lose tangential friction (we substitute rigid contact with rubber’s μ where it matters), and loaded articulations cannot receive new collision geometry (we place task geometry on objects instead). The optical/audio channel of the click is not modelled—the click is kinematic. The insertion by the robot (rather than by an applied-force protocol) is future work, as is the end-to-end task from rest with a verifier verdict.

Outlook. The pen cap is a stand-in for an enormous class of everyday snap-fit interactions—food containers, battery covers, cable clips, medicine bottles—whose simulation is usually faked by pose checks. The micro-law catalogue and the probe-first doctrine transfer directly; the measured force signatures give reinforcement learning and imitation pipelines an honest reward surface where “closed” means *clicked, seated, and retained*.

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